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Adaptive connection-based graphical user interfaces for video conferences

Abstract

A video conference that is in progress is run through a device, such as a portable device. Video of the video conference is displayed on a display of the device. The device detects when a connection is made between the device and an external display during the video conference that is in progress. When the device detects the connection between the device and the external display, the device causes the video of the video conference to be displayed on the external display. When the video of the video conference is displayed on the external display, a control panel is displayed on the display of the device. The device retains the connection to the video conference while the video of the video conference is displayed on the external display and the control panel is displayed on the display of the device.

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Background/Summary

FIELD

(1) This disclosure generally relates to adaptive graphical user interfaces (GUIs) for video conferences, and in particular, UIs that are adaptive based on a connection to one or more devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

(2) FIG. 1 is a block diagram of an example of an electronic computing and communications system.

(3) FIG. 2 is a block diagram of an example internal configuration of a computing device of an

electronic computing and communications system.

(4) FIG. 3 is a block diagram of an example of a software platform implemented by an electronic computing and communications system.

(5) FIG. 4 is a block diagram of an example of a system for adaptive connection-based GUIs for video conferences.

(6) FIG. 5 is a block diagram of an example of a device configured for adaptive connection-based GUIs for video conferences.

(7) FIG. 6A is a diagram of an example of a display of a portable device that is configured for adaptive connection-based GUIs for video conferences prior to connection to an external device.

(8) FIG. 6B is a diagram of an example of a display of the portable device of FIG. 6A when the portable device is connected to an external device.

(9) FIG. 7 is a flowchart of an example of a method for generating adaptive connection-based GUIs for video conferences.

DETAILED DESCRIPTION

(10) Enterprise entities rely upon several modes of communication to support their operations, including telephone, email, internal messaging, and the like. These separate modes of communication have historically been implemented by service providers whose services are not integrated with one another. The disconnect between these services, in at least some cases, requires information to be manually passed by users from one service to the next. Furthermore, some services, such as telephony services, are traditionally delivered via on-premises systems, meaning that remote workers and those who are generally increasingly mobile may be unable to rely upon them. One type of system which addresses problems such as these includes a unified communications as a service (UCaaS) platform, which includes several communications services integrated over a network, such as the Internet, to deliver a complete communication experience regardless of physical location.

(11) Software platforms, such as UCaaS platforms, can implement software services enabling communications over one or more modalities such as telephony, email, text messaging, chat, or audio or video conferencing. A modality, such as video conferencing, may be accessed using devices in communication with the software platform. Example devices that can be used to access video conferencing can include portable devices, such as mobile phones, tablet computers, and laptop computers. In some examples, video conferencing can be accessed using other devices, such as desktop computers or conference room devices.

(12) Conference participants may often be connected to a video conference via a portable device when the need arises to switch the display of the content of the video conference from the portable device to a conference room device, such as an external monitor, while retaining the conference connection through the portable device. Typical solutions allow for the content of the video conference to be displayed on a conference room device; however, in such cases, the display is mirrored such that the portable device display and the conference room device both display identical content, resulting in an inefficient use of the available display area. Some current solutions require ultrasonic conference room equipment to detect the portable device and switch the video conference session from the portable device to the conference room device. Moving the video conference session from the portable device to the conference room device can be a significant drain on computing resources and introduces the risk of dropping the video conference session while same is being transferred between the subject devices. In addition, the ultrasonic conference room equipment can be costly to purchase and maintain, as it requires pre-configuration by information technology (IT) staff.

(13) Implementations of this disclosure address problems such as these by adapting connection-based UIs for video conferences. A client application running on a portable device detects that an external display is connected to the portable device. Based on the detection, the client application transfers the GUI within which the media of the video conference is rendered to the external

display and switches the visual output on the portable device to an adaptive GUI, such as a controller interface configured to control one or more functions of the video conference media output to the external display. The connection to the video conference is retained on the portable device, such that the video conference session is not moved to the external display or another conference room device. The adaptive GUI can be configured based on one or more user preferences, such as to customize GUI elements according to the individual user. In some examples, the adaptive GUI can be configured using a machine learning (ML) model that is trained using a data set of display preferences from previous instances of when any user connected any user device to an external display.

(14) To describe some implementations in greater detail, reference is first made to examples of hardware and software structures used to implement adaptive connection-based GUIs for video conferences. FIG. 1 is a block diagram of an example of an electronic computing and communications system **100**, which can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.

(15) The system **100** includes one or more customers, such as customers **102A** through **102B**, which may each be a public entity, private entity, or another corporate entity or individual that purchases or otherwise uses software services, such as of a UCaaS platform provider. Each customer can include one or more clients. For example, as shown and without limitation, the customer **102A** can include clients **104A** through **104B**, and the customer **102B** can include clients **104C** through **104D**. A customer can include a customer network or domain. For example, and without limitation, the clients **104A** through **104B** can be associated or communicate with a customer network or domain for the customer **102A** and the clients **104C** through **104D** can be associated or communicate with a customer network or domain for the customer **102B**.

(16) A client, such as one of the clients **104A** through **104D**, may be or otherwise refer to one or both of a client device or a client application. Where a client is or refers to a client device, the client can comprise a computing system, which can include one or more computing devices, such as a mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices. Where a client instead is or refers to a client application, the client can be an instance of software running on a customer device (e.g., a client device or another device). In some implementations, a client can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients.

(17) The system **100** can include a number of customers and/or clients or can have a configuration of customers or clients different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include hundreds or thousands of customers, and at least some of the customers can include or be associated with a number of clients.

(18) The system **100** includes a datacenter **106**, which may include one or more servers. The datacenter **106** can represent a geographic location, which can include a facility, where the one or more servers are located. The system **100** can include a number of datacenters and servers or can include a configuration of datacenters and servers different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include tens of datacenters, and at least some of the datacenters can include hundreds or another suitable number of servers. In some implementations, the datacenter **106** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than the customer domains for the customers **102A** through **102B**.

(19) The datacenter **106** includes servers used for implementing software services of a UCaaS platform. The datacenter **106** as generally illustrated includes an application server **108**, a database server **110**, and a telephony server **112**. The servers **108** through **112** can each be a computing system, which can include one or more computing devices, such as a desktop computer, a server

computer, or another computer capable of operating as a server, or a combination thereof. A suitable number of each of the servers **108** through **112** can be implemented at the datacenter **106**. The UCaaS platform uses a multi-tenant architecture in which installations or instantiations of the servers **108** through **112** is shared amongst the customers **102A** through **102B**.

(20) In some implementations, one or more of the servers **108** through **112** can be a non-hardware server implemented on a physical device, such as a hardware server. In some implementations, a combination of two or more of the application server **108**, the database server **110**, and the telephony server **112** can be implemented as a single hardware server or as a single non-hardware server implemented on a single hardware server. In some implementations, the datacenter **106** can include servers other than or in addition to the servers **108** through **112**, for example, a media server, a proxy server, or a web server.

(21) The application server **108** runs web-based software services deliverable to a client, such as one of the clients **104A** through **104D**. As described above, the software services may be of a UCaaS platform. For example, the application server **108** can implement all or a portion of a UCaaS platform, including conferencing software, messaging software, and/or other intra-party or inter-party communications software. The application server **108** may, for example, be or include a unitary Java Virtual Machine (JVM).

(22) In some implementations, the application server **108** can include an application node, which can be a process executed on the application server **108**. For example, and without limitation, the application node can be executed in order to deliver software services to a client, such as one of the clients **104A** through **104D**, as part of a software application. The application node can be implemented using processing threads, virtual machine instantiations, or other computing features of the application server **108**. In some such implementations, the application server **108** can include a suitable number of application nodes, depending upon a system load or other characteristics associated with the application server **108**. For example, and without limitation, the application server **108** can include two or more nodes forming a node cluster. In some such implementations, the application nodes implemented on a single application server **108** can run on different hardware servers.

(23) The database server **110** stores, manages, or otherwise provides data for delivering software services of the application server **108** to a client, such as one of the clients **104A** through **104D**. In particular, the database server **110** may implement one or more databases, tables, or other information sources suitable for use with a software application implemented using the application server **108**. The database server **110** may include a data storage unit accessible by software executed on the application server **108**. A database implemented by the database server **110** may be a relational database management system (RDBMS), an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. The system **100** can include one or more database servers, in which each database server can include one, two, three, or another suitable number of databases configured as or comprising a suitable database type or combination thereof.

(24) In some implementations, one or more databases, tables, other suitable information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the system **100** other than the database server **110**, for example, the client **104** or the application server **108**.

(25) The telephony server **112** enables network-based telephony and web communications from and to clients of a customer, such as the clients **104A** through **104B** for the customer **102A** or the clients **104C** through **104D** for the customer **102B**. Some or all of the clients **104A** through **104D** may be voice over internet protocol (VOIP)-enabled devices configured to send and receive calls over a network **114**. In particular, the telephony server **112** includes a session initiation protocol (SIP) zone and a web zone. The SIP zone enables a client of a customer, such as the customer **102A**

or **102B**, to send and receive calls over the network **114** using SIP requests and responses. The web zone integrates telephony data with the application server **108** to enable telephony-based traffic access to software services run by the application server **108**. Given the combined functionality of the SIP zone and the web zone, the telephony server **112** may be or include a cloud-based private branch exchange (PBX) system.

(26) The SIP zone receives telephony traffic from a client of a customer and directs same to a destination device. The SIP zone may include one or more call switches for routing the telephony traffic. For example, to route a VOIP call from a first VOIP-enabled client of a customer to a second VOIP-enabled client of the same customer, the telephony server **112** may initiate a SIP transaction between a first client and the second client using a PBX for the customer. However, in another example, to route a VOIP call from a VOIP-enabled client of a customer to a client or non-client device (e.g., a desktop phone which is not configured for VOIP communication) which is not VOIP-enabled, the telephony server **112** may initiate a SIP transaction via a VOIP gateway that transmits the SIP signal to a public switched telephone network (PSTN) system for outbound communication to the non-VOIP-enabled client or non-client phone. Hence, the telephony server **112** may include a PSTN system and may in some cases access an external PSTN system.

(27) The telephony server **112** includes one or more session border controllers (SBCs) for interfacing the SIP zone with one or more aspects external to the telephony server **112**. In particular, an SBC can act as an intermediary to transmit and receive SIP requests and responses between clients or non-client devices of a given customer with clients or non-client devices external to that customer. When incoming telephony traffic for delivery to a client of a customer, such as one of the clients **104A** through **104D**, originating from outside the telephony server **112** is received, a SBC receives the traffic and forwards it to a call switch for routing to the client.

(28) In some implementations, the telephony server **112**, via the SIP zone, may enable one or more forms of peering to a carrier or customer premise. For example, Internet peering to a customer premise may be enabled to ease the migration of the customer from a legacy provider to a service provider operating the telephony server **112**. In another example, private peering to a customer premise may be enabled to leverage a private connection terminating at one end at the telephony server **112** and at the other end at a computing aspect of the customer environment. In yet another example, carrier peering may be enabled to leverage a connection of a peered carrier to the telephony server **112**.

(29) In some such implementations, a SBC or telephony gateway within the customer environment may operate as an intermediary between the SBC of the telephony server **112** and a PSTN for a peered carrier. When an external SBC is first registered with the telephony server **112**, a call from a client can be routed through the SBC to a load balancer of the SIP zone, which directs the traffic to a call switch of the telephony server **112**. Thereafter, the SBC may be configured to communicate directly with the call switch.

(30) The web zone receives telephony traffic from a client of a customer, via the SIP zone, and directs same to the application server **108** via one or more Domain Name System (DNS) resolutions. For example, a first DNS within the web zone may process a request received via the SIP zone and then deliver the processed request to a web service which connects to a second DNS at or otherwise associated with the application server **108**. Once the second DNS resolves the request, it is delivered to the destination service at the application server **108**. The web zone may also include a database for authenticating access to a software application for telephony traffic processed within the SIP zone, for example, a softphone.

(31) The clients **104A** through **104D** communicate with the servers **108** through **112** of the datacenter **106** via the network **114**. The network **114** can be or include, for example, the Internet, a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client and one or more servers. In some implementations, a client can connect to the

network **114** via a communal connection point, link, or path, or using a distinct connection point, link, or path. For example, a connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

(32) The network **114**, the datacenter **106**, or another element, or combination of elements, of the system **100** can include network hardware such as routers, switches, other network devices, or combinations thereof. For example, the datacenter **106** can include a load balancer **116** for routing traffic from the network **114** to various servers associated with the datacenter **106**. The load balancer **116** can route, or direct, computing communications traffic, such as signals or messages, to respective elements of the datacenter **106**.

(33) For example, the load balancer **116** can operate as a proxy, or reverse proxy, for a service, such as a service provided to one or more remote clients, such as one or more of the clients **104A** through **104D**, by the application server **108**, the telephony server **112**, and/or another server. Routing functions of the load balancer **116** can be configured directly or via a DNS. The load balancer **116** can coordinate requests from remote clients and can simplify client access by masking the internal configuration of the datacenter **106** from the remote clients.

(34) In some implementations, the load balancer **116** can operate as a firewall, allowing or preventing communications based on configuration settings. Although the load balancer **116** is depicted in FIG. 1 as being within the datacenter **106**, in some implementations, the load balancer **116** can instead be located outside of the datacenter **106**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both within and outside of the datacenter **106**. In some implementations, the load balancer **116** can be omitted.

(35) FIG. 2 is a block diagram of an example internal configuration of a computing device **200** of an electronic computing and communications system. In one configuration, the computing device **200** may implement one or more of the client **104**, the application server **108**, the database server **110**, or the telephony server **112** of the system **100** shown in FIG. 1.

(36) The computing device **200** includes components or units, such as a processor **202**, a memory **204**, a bus **206**, a power source **208**, peripherals **210**, a user interface **212**, a network interface **214**, other suitable components, or a combination thereof. One or more of the memory **204**, the power source **208**, the peripherals **210**, the user interface **212**, or the network interface **214** can communicate with the processor **202** via the bus **206**.

(37) The processor **202** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **202** can include another type of device, or multiple devices, configured for manipulating or processing information. For example, the processor **202** can include multiple processors interconnected in one or more manners, including hardwired or networked. The operations of the processor **202** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **202** can include a cache, or cache memory, for local storage of operating data or instructions.

(38) The memory **204** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory can be random access memory (RAM) (e.g., a DRAM module, such as DDR SDRAM). In another example, the non-volatile memory of the memory **204** can be a disk drive, a solid state drive, flash memory, or phase-change memory. In some implementations, the memory **204** can be distributed across multiple devices. For example, the memory **204** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices.

(39) The memory **204** can include data for immediate access by the processor **202**. For example, the memory **204** can include executable instructions **216**, application data **218**, and an operating system **220**. The executable instructions **216** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be

executed by the processor **202**. For example, the executable instructions **216** can include instructions for performing some or all of the techniques of this disclosure. The application data **218** can include user data, database data (e.g., database catalogs or dictionaries), or the like. In some implementations, the application data **218** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof. The operating system **220** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®, an operating system for a mobile device, such as a smartphone or tablet device; or an operating system for a non-mobile device, such as a mainframe computer.

(40) The power source **208** provides power to the computing device **200**. For example, the power source **208** can be an interface to an external power distribution system. In another example, the power source **208** can be a battery, such as where the computing device **200** is a mobile device or is otherwise configured to operate independently of an external power distribution system. In some implementations, the computing device **200** may include or otherwise use multiple power sources. In some such implementations, the power source **208** can be a backup battery.

(41) The peripherals **210** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **200** or the environment around the computing device **200**. For example, the peripherals **210** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **200**, such as the processor **202**. In some implementations, the computing device **200** can omit the peripherals **210**.

(42) The user interface **212** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen, or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, or other suitable display.

(43) The network interface **214** provides a connection or link to a network (e.g., the network **114** shown in FIG. 1). The network interface **214** can be a wired network interface or a wireless network interface. The computing device **200** can communicate with other devices via the network interface **214** using one or more network protocols, such as using Ethernet, transmission control protocol (TCP), internet protocol (IP), power line communication, an IEEE 802.X protocol (e.g., Wi-Fi, Bluetooth, or ZigBee), infrared, visible light, general packet radio service (GPRS), global system for mobile communications (GSM), code-division multiple access (CDMA), Z-Wave, another protocol, or a combination thereof.

(44) FIG. 3 is a block diagram of an example of a software platform **300** implemented by an electronic computing and communications system, for example, the system **100** shown in FIG. 1. The software platform **300** is a UCaaS platform accessible by clients of a customer of a UCaaS platform provider, for example, the clients **104A** through **104B** of the customer **102A** or the clients **104C** through **104D** of the customer **102B** shown in FIG. 1. The software platform **300** may be a multi-tenant platform instantiated using one or more servers at one or more datacenters including, for example, the application server **108**, the database server **110**, and the telephony server **112** of the datacenter **106** shown in FIG. 1.

(45) The software platform **300** includes software services accessible using one or more clients. For example, a customer **302** as shown includes four clients—a desk phone **304**, a computer **306**, a mobile device **308**, and a shared device **310**. The desk phone **304** is a desktop unit configured to at least send and receive calls and includes an input device for receiving a telephone number or extension to dial to and an output device for outputting audio and/or video for a call in progress. The computer **306** is a desktop, laptop, or tablet computer including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The mobile device **308** is a smartphone, wearable device, or other mobile computing aspect including an input device for receiving some form of user input and an output device for outputting

information in an audio and/or visual format. The desk phone **304**, the computer **306**, and the mobile device **308** may generally be considered personal devices configured for use by a single user. The shared device **310** is a desk phone, a computer, a mobile device, or a different device which may instead be configured for use by multiple specified or unspecified users.

(46) Each of the clients **304** through **310** includes or runs on a computing device configured to access at least a portion of the software platform **300**. In some implementations, the customer **302** may include additional clients not shown. For example, the customer **302** may include multiple clients of one or more client types (e.g., multiple desk phones or multiple computers) and/or one or more clients of a client type not shown in FIG. 3 (e.g., wearable devices or televisions other than as shared devices). For example, the customer **302** may have tens or hundreds of desk phones, computers, mobile devices, and/or shared devices.

(47) The software services of the software platform **300** generally relate to communications tools, but are in no way limited in scope. As shown, the software services of the software platform **300** include telephony software **312**, conferencing software **314**, messaging software **316**, and other software **318**. Some or all of the software **312** through **318** uses customer configurations **320** specific to the customer **302**. The customer configurations **320** may, for example, be data stored within a database or other data store at a database server, such as the database server **110** shown in FIG. 1.

(48) The telephony software **312** enables telephony traffic between ones of the clients **304** through **310** and other telephony-enabled devices, which may be other ones of the clients **304** through **310**, other VOIP-enabled clients of the customer **302**, non-VOIP-enabled devices of the customer **302**, VOIP-enabled clients of another customer, non-VOIP-enabled devices of another customer, or other VOIP-enabled clients or non-VOIP-enabled devices. Calls sent or received using the telephony software **312** may, for example, be sent or received using the desk phone **304**, a softphone running on the computer **306**, a mobile application running on the mobile device **308**, or using the shared device **310** that includes telephony features.

(49) The telephony software **312** further enables phones that do not include a client application to connect to other software services of the software platform **300**. For example, the telephony software **312** may receive and process calls from phones not associated with the customer **302** to route that telephony traffic to one or more of the conferencing software **314**, the messaging software **316**, or the other software **318**.

(50) The conferencing software **314** enables audio, video, and/or other forms of conferences between multiple participants, such as to facilitate a conference between those participants. In some cases, the participants may all be physically present within a single location, for example, a conference room, in which the conferencing software **314** may facilitate a conference between only those participants and using one or more clients within the conference room. In some cases, one or more participants may be physically present within a single location and one or more other participants may be remote, in which the conferencing software **314** may facilitate a conference between all of those participants using one or more clients within the conference room and one or more remote clients. In some cases, the participants may all be remote, in which the conferencing software **314** may facilitate a conference between the participants using different clients for the participants. The conferencing software **314** can include functionality for hosting, presenting scheduling, joining, or otherwise participating in a conference. The conferencing software **314** may further include functionality for recording some or all of a conference and/or documenting a transcript for the conference.

(51) The messaging software **316** enables instant messaging, unified messaging, and other types of messaging communications between multiple devices, such as to facilitate a chat or other virtual conversation between users of those devices. The unified messaging functionality of the messaging software **316** may, for example, refer to email messaging which includes a voicemail transcription service delivered in email format.

(52) The other software **318** enables other functionality of the software platform **300**. Examples of the other software **318** include, but are not limited to, device management software, resource provisioning and deployment software, administrative software, third party integration software, and the like. In one particular example, the other software **318** enables adaptive connection-based GUIs for video conferences.

(53) The software **312** through **318** may be implemented using one or more servers, for example, of a datacenter such as the datacenter **106** shown in FIG. **1**. For example, one or more of the software **312** through **318** may be implemented using an application server, a database server, and/or a telephony server, such as the servers **108** through **112** shown in FIG. **1**. In another example, one or more of the software **312** through **318** may be implemented using servers not shown in FIG. **1**, for example, a meeting server, a web server, or another server. In yet another example, one or more of the software **312** through **318** may be implemented using one or more of the servers **108** through **112** and one or more other servers. The software **312** through **318** may be implemented by different servers or by the same server.

(54) Features of the software services of the software platform **300** may be integrated with one another to provide a unified experience for users. For example, the messaging software **316** may include a user interface element configured to initiate a call with another user of the customer **302**. In another example, the telephony software **312** may include functionality for elevating a telephone call to a conference. In yet another example, the conferencing software **314** may include functionality for sending and receiving instant messages between participants and/or other users of the customer **302**. In yet another example, the conferencing software **314** may include functionality for file sharing between participants and/or other users of the customer **302**. In some implementations, some or all of the software **312** through **318** may be combined into a single software application run on clients of the customer, such as one or more of the clients **304** through **310**.

(55) FIG. **4** is a block diagram of an example of a system **400** for adaptive connection-based GUIs for video conferences. The system **400** includes a user device **402**, a display device **404**, a server **406**, and one or more user devices **408A-N**. In some implementations, the system **400** may include more than one display device. The user device **402** may be a device such as the computer **306** or the mobile device **308** shown in FIG. **3**. The server **406** may be implemented as the software platform **300** shown in FIG. **3**. The one or more user devices **408A-N** may each be one of the clients **304** through **310** shown in FIG. **3**.

(56) The user device **402** includes a processor **410**, a memory **412**, a connector **414**, a display **416**, a camera **418**, a microphone **420**, and a speaker **422**. The processor **410** is configured to run a client application stored on the memory **412** to connect to a video conference via the server **406**. The processor **410** is configured to generate a GUI within the client application that includes video content of the video conference and display the GUI on the display **416**. The GUI may be a customizable GUI such that it can display a gallery view of conference participants, a speaker view of one or more conference participants, a shared content view (e.g., a presentation or an interactive whiteboard), a chat room view, a transcription view, or another view of the video conference. The display **416** may be a touch display that is configured to obtain user input via a touch or a gesture. The camera **418** is configured to obtain video for the video conference. The microphone **420** is configured to obtain audio for the video conference. The speaker **422** is configured to output audio obtained from conference participant devices, such as the user devices **408A-N**.

(57) The processor **410** is configured to execute instructions stored in the memory **412** to cause the user device **402** to detect a connection between the user device **402** and the display device **404**. Software executed on the processor **410** is configured to monitor the connector **414** to detect a connection when a device, such as the device **404**, or a connector thereof, is inserted into the connector **414** or otherwise connected to the connector **414**. When a connection is detected, the software executed on the processor **410** is configured to exchange information with the device **404**.

to perform an action, such as configure one or more settings on the device **404**, for example. The one or more settings may include display settings, such as a resolution setting, a refresh rate setting, a display aspect ratio setting, or another display setting. The display device **404** is not connected to the video conference and may be an external monitor, a projector, a television, or another display that can be connected to the user device **402**. The processor **410** may execute the instructions stored in the memory **412** to cause the user device **402** to detect the connection via the connector **414** using the client application. In one example, the connection may be detected prior to joining the video conference. In another example, the connection may be detected during the video conference. The connector **414** may be a wired connector, such as a universal serial bus (USB) connector or another type of connector. In some implementations, the connector **414** may enable a wireless connection, for example, using near-field communication (NFC), Bluetooth, Wi-Fi, or another wireless connection.

(58) When the software executed using the processor **410** detects the connection between the user device **402** and the display device **404**, the software transfers the display of the GUI to the display device **404**. Transferring the display of the GUI to the display device **404** includes transmitting data configured to cause a rendering of the GUI from the device **402** to the display device **404**. In some implementations, the processor **410** may be configured to execute instructions stored on the memory **412** to render the GUI based on the display capabilities of the display device **404** prior to transferring the GUI for display on the display device **404**. For example, the user device **402** can be configured to detect the display capabilities of the display device **404** and generate or obtain a second GUI based on the display capabilities of the display device **404** while continuing to output the GUI at the user device **402** such that the second GUI is ready to be transferred to the display device **404**. Obtaining the second GUI may include accessing the second GUI from a local data store, such as memory **412**, or remote data store, such as a data store on server **406**. When the user device **402** is connected to a video conference that is in progress, the software executed using the processor **410** transfers the display of the GUI to the display device **404** and retains the connection to the video conference on the user device **402**. The processor **410** is configured to execute instructions stored on the memory **412** to generate or obtain the second GUI based on the detection of the connection between the user device **402** and the display device **404** and display the second GUI on the display **416**. In some implementations, the second GUI may be a customizable GUI that can include a controller to control functions of the video conference, a gallery view of conference participants, a speaker view of one or more conference participants, or an interactive whiteboard. When the second GUI is displayed on the display **416**, the processor **410** continues to display the GUI on the display device **404**.

(59) The user device **402** uses the adaptive connection-based GUIs to display different video conference information on multiple displays when one or more display devices, such as display device **404**, are connected to the user device **402**. The video conference is run though the user device **402**. When the display device **404** is connected to the user device **402**, a first GUI being displayed on the user device is transferred to the display device **404**, and the display **416** of the user device **402** is updated to display a second GUI associated with the video conference. When the first GUI is being displayed on the display device **404** and the second GUI is displayed on the user device **402**, the user device **402** retains the connection to the video conference.

(60) FIG. 5 is a block diagram of an example of a device **500** configured for adaptive connection-based GUIs for video conferences. The device **500** may, for example, be the user device **402** shown in FIG. 4. In this example, the device **500** is connected to a display device **502**. The display device **502** may, for example, be the display device **404** shown in FIG. 4. The device **500** includes a display **504**, which may, for example, be the display **416** shown in FIG. 4.

(61) The device **500** includes a client application **506** that is stored in a memory, such as the memory **412** shown in FIG. 4 and executed by a processor, such as the processor **410** shown in FIG. 4. For example, the client application **506** may be software usable to access software services

of a software platform (e.g., a UCaaS platform) from the device **500**. The client application **506** includes connection detection software **508** and video conference software **510**.

(62) The connection detection software **508** is configured to detect when the display device **502** is connected to the device **500**. The connection detection software **508** is configured to monitor a connector, such as the connector **414** shown in FIG. **4**, to detect a connection when a device, such as the display device **502**, or a connector thereof, is inserted into the connector **414** or otherwise connected to the connector **414**. When a connection is detected, the connection detection software **508** is configured to exchange information with the display device **502** to perform an action, such as configure one or more settings on the display device **502**, for example. The one or more settings may include display settings, such as a resolution setting, a refresh rate setting, a display aspect ratio setting, or another display setting. In addition, the connection detection software **508** is configured to detect when the display device **502** is disconnected from the device **500**. When the display device **502**, or a connector thereof, is removed from the connector **414** or the connection is otherwise terminated, the connection detection software determines that the display device **502** is disconnected from the device **500**. When the connection detection software **508** no longer detects a connection, the device **500** is configured to perform an action, such as configure one or more settings on a display of the device **500**, for example.

(63) The video conference software **510** is client-side software that is configured to communicate with server-side software that facilitates a video conference (e.g., the conferencing software **314** of FIG. **3**) to connect the device **500** to the video conference. The video conference software **510** is configured to process audio data of the conference such that the audio of the conference is output via a speaker of the device **500**, such as the speaker **422** shown in FIG. **4**. The video conference software **510** is configured to process video data of the conference to generate or obtain a GUI **512** and a GUI **514**. The GUI **512** and the GUI **514** may be generated based on a detection of a connection between the device **500** and the display device **502** by the connection detection software **508**. If multiple display devices are connected to the device **500**, the video conference software **510** is configured to generate a GUI for each connected display device. The GUI **512** and the GUI **514** may form a switchable UI that is configured to switch between the GUI **512** and the GUI **514** based on a detected connection between the device **500** and the display device **502** or a detected disconnection of the display device **502** from the device **500**. The GUI **512** and the GUI **514** are output via a display, such as the display **504**, the display device **502**, or both.

(64) The GUI **512** and the GUI **514** may be configurable, for example, based on a user preference. The GUI **512** and the GUI **514** can be configured to have different GUI information based on the user preference in order to render different displays as described in the example use cases below. In some implementations, the user preference may be learned using an ML model. The ML model may be trained using a data set of display preferences of previous instances of when a user connected any device, such as the device **500** to any display, such as the display device **502**. The ML model may be updated each time the device **500** is connected to a display. The ML model may determine the GUI configurations based on the display device type to which the device **500** is connected. The ML model may base the GUI configurations on one or more display device capabilities. The one or more display device capabilities may include a display size, a display aspect ratio, a display resolution, a display refresh rate, or another device capability. In an example, the ML model may determine that when the device **500** is connected to the display device **502**, the user prefers to display a GUI showing conference participants in a gallery view on the display device **502** and a GUI showing conference controls on the device **500**.

(65) The following example use case is provided as an illustrative implementation of the device **500** shown in FIG. **5**. In this example, the device **500** may be a portable device, such as a mobile phone or a tablet computer. The device **500** is connected to a video conference via the video conference software **510**. The video conference software **510** is configured to output the GUI **512** for display on the display **504**. In this example, the GUI **512** is configured to include GUI

information for rendering conference participants on the display **504**. At some point during the conference, the connection detection software **508** detects that display device **502** is connected to the device **500**. In this example, the display device **502** may be an external display. Based on the detected connection between the display device **502** and the device **500**, the video conference software **510** is configured to output the GUI **512** for display on the display device **502** such that the rendering of the conference participants continues on the display device **502**. While the GUI **512** is being displayed on the device **502**, the video conference software **510** is configured to output the GUI **514** for display on the display **504**. In this example, the GUI **514** is configured to include GUI information for rendering a control panel that includes one or more controls for the video conference. While the GUI **512** is displayed on the display device **502** and the GUI **514** is displayed on the display **504**, the video conference software **510** is configured to retain the connection to the video conference via the device **500**.

(66) At some point during the conference, the connection detection software **508** detects that the display device **502** is disconnected from the device **500**. Based on the detected disconnection of the display device **502** from the device **500**, the video conference software **510** is configured terminate the output of the GUI **512** to the device **502**, terminate the output of the GUI **514** to the display **504**, and output the GUI **512** for display on the display **504** such that the rendering of the conference participants continues on the display **504** of the device **500**. While the GUI **512** is displayed on the display **504**, the video conference software **510** is configured to retain the connection to the video conference via the device **500**.

(67) In a second example use case, the device **500** may be a portable device, such as a mobile phone or a tablet computer that is connected to a video conference via the video conference software **510**. In this example, the video conference software **510** is configured to output the GUI **512** for display on the display device **502** and output the GUI **514** for display on the display **504**. The display device **502** may be an external display, such as a projector in a conference room. In this example, the GUI **512** is configured to include GUI information for rendering conference participants on the display device **502** and the GUI **514** is configured to include GUI information for rendering a control panel on the display **504**. The control panel includes controls for the video conference.

(68) At some point during the conference, the video conference software **510** receives an input via the control panel. The input may be a touch input or a gesture input that indicates a command to change the GUI that is displayed on the display device **502**, the display **504**, or both. In this example, the command may indicate to display a whiteboard on the display device **502** and conference participants on the display **504**. Based on the command, the video conference software **510** is configured to output the GUI **512** for display on the display **504** such that the rendering of the conference participants continues on the display **504**.

(69) While the GUI **512** is being displayed on the display **504**, the video conference software **510** is configured to output the GUI **514** for display on the display device **502**. In this example, the GUI **514** is configured to include GUI information for rendering a whiteboard of the video conference. In some implementations, the video conference software **510** may be configured to hide the control panel so that it can be accessed via a touch input (e.g., tapping on an icon on the display **504**) or a gesture input (e.g., swiping on the display **504**). While the GUI **512** is displayed on the display **504** and the GUI **514** is displayed on the display device **502**, the video conference software **510** is configured to retain the connection to the video conference via the device **500**. Although this example use case describes rendering a whiteboard of the video conference on the display device **502** and rendering conference participants on the display **504**, other information can be rendered on the displays. For example, the video conference software **510** may be configured to display a participant (e.g., a speaker) on the display device **502** and conference participants on the display **504** based on the command.

(70) In a third example use case, the device **500** may be a laptop computer or a desktop computer

that is connected to multiple external monitors. In this example, the device **500** is connected to three external monitors and configured to connect to a video conference via the video conference software **510**. The video conference software **510** is configured to output the GUI **512** for display on the first external monitor, output the GUI **514** for display on the second external monitor, and output a third graphical UI for display on the third external monitor. In this example, the GUI **512** is configured to include GUI information for rendering conference participants on the first external monitor, the GUI **514** is configured to include GUI information for rendering a presentation on the second external monitor, and the third GUI is configured to include GUI information for rendering a whiteboard on the third external monitor. The configuration of the information displayed on each external monitor may be based on a user preference. In an example, the user preference may be learned using an ML model. The ML model may be trained using a data set of display preferences of previous instances of when a user connected the device **500** to multiple external monitors. In some implementations, the user preference may be based on the type of video conference to which the device **500** is connected. For example, a first user preference may be used for a webinar, a second user preference may be used for a small video conference (e.g., less than 10 participants), and a third user preference may be used for a large video conference (e.g., greater than 10 participants).

(71) In a webinar example, the user preference may indicate to display a speaker on the first external monitor, a presentation on the second external monitor, and participant comments and/or questions on the third external monitor. In a small video conference example, the user preference may indicate to display a speaker on the first external monitor, a whiteboard on the second external monitor, and a chat room on the third external monitor. In a large video conference example, the user preference may indicate to display a speaker on the first external monitor and to display the other participants spread across the second and third external monitors.

(72) FIG. **6A** is a diagram of an example of a display **600** of a portable device **602** that is configured for adaptive connection-based GUIs for video conferences prior to connection to an external device **604**. FIG. **6B** is a diagram of an example of the display **600** of the portable device **602** shown in FIG. **6A** when the portable device **602** is connected to the external device **604**. The portable device **602** may, for example, be the device **500** shown in FIG. **5**. The external device **604** may, for example, be the display device **502** shown in FIG. **5**.

(73) The portable device **602** is connected to a video conference. The display **600** includes a GUI **606**. The GUI **606** is configurable and can display information based on a user preference. In this example, the GUI **606** is configured to display conference participants prior to a connection between the portable device **602** and the external device **604**. In other examples, the GUI **606** can be configured to display a participant that is speaking in the video conference, a presentation, a whiteboard, a chat room, one or more conference controls, or another visual output associated with the video conference. In some examples, the GUI **606** can be configured to display various combinations of a participant that is speaking in the video conference, a presentation, a whiteboard, a chat room, one or more conference controls, or another visual output associated with the video conference in one or more portions of the GUI **606**.

(74) As shown in FIG. **6B**, when the portable device **602** is connected to the external device **604**, the information from the GUI **606** is transferred to a display **608** of the external device **604** such that the display of the conference participants continues on the display **608** of the external device **604**. While the GUI **606** displayed on the display **608** of the external device **604**, the display **600** of the portable device **602** is configured to display an updated GUI **610**. The updated GUI **610** is configurable and can display information based on a user preference. In this example, the updated GUI **610** is configured to display a control panel to control one or more functions of the conference. In other examples, the updated GUI **610** can be configured to display a participant that is speaking in the video conference, a presentation, a whiteboard, a chat room, one or more conference controls, or another visual output associated with the video conference. In some

examples, the updated GUI **610** can be configured to display various combinations of a participant that is speaking in the video conference, a presentation, a whiteboard, a chat room, one or more conference controls, or another visual output associated with the video conference in one or more portions of the updated GUI **610**.

(75) The control panel can be configured to include one or more buttons or sliders to control the one or more functions of the conference. For example, the control panel may include a mute button that is configured to mute or unmute a microphone of the portable device **602**. The control panel can be configured to include a video button that is used to start or stop a camera of the portable device. In some examples, the control panel can be customized to include a share button configured to enable sharing of content, such as a presentation, a screen share, or other content, from the portable device **602** and/or a security button to enable or disable one or more features for the conference. Example features that can be enabled via the security button include locking the conference to prevent other participants from joining the conference in progress, enabling or disabling a waiting room for the conference, enabling or disabling screen share capabilities for participants, enabling or disabling chat capabilities for participants, enabling or disabling participants to unmute themselves, or enabling or disabling participants to annotate a shared screen. A view button can be configured to be displayed on the control panel where the view button is configured to change a layout of the GUI **606** on the external device **604** and/or the GUI **610** on the portable device **602**. Some configurations of the control panel may include a camera control button that is configured to change a camera setting, a camera preset, or switch to a different camera, a participants button that is configured to enable viewing and management of the conference participants, a record button that is configured to start and stop recording of the conference, and/or a chat button that is configured to access an in-conference chat room. One or more control panel configurations may include a settings button configured to access conference settings, a volume slider that is configured to adjust a speaker volume of the portable device **602**, and/or an exit button that is configured to end the video conference.

(76) To further describe some implementations in greater detail, reference is next made to examples of techniques which may be performed by or using a device for generating adaptive connection-based GUIs for video conferences. FIG. 7 is a flowchart of an example of a method **700** for generating adaptive connection-based GUIs for video conferences. The method **700** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. 1-6B. The method **700** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the method **700** or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

(77) For simplicity of explanation, the method **700** is depicted and described herein as a series of steps or operations. However, the steps or operations in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

(78) At **702**, the method **700** includes displaying a first GUI within a client application running at a portable device. The portable device is connected to a video conference. The first GUI includes video content of the video conference. Audio for the video conference may be obtained via a microphone of the portable device. Video for the video conference may be obtained via a camera of the portable device.

(79) At **704**, the method **700** includes detecting a connection between the portable device and a display device. The display device may be an external display that is not connected to the video conference. The detected connection may be a wired connection or a wireless connection. For

example, a wired connection may be via a USB connector (e.g., USB-A, USB-B, or USB-C), a high definition multimedia interface (HDMI) connector, a video graphics array (VGA) connector, a digital visual interface (DVI) connector, a display port connector, or another type of connector. Example wireless connections include NFC, Bluetooth, Wi-Fi, or other wireless connections.

(80) At **706**, the method **700** includes, based on the detected connection, causing a further display of the first GUI at the display device. The client application causes the further display of the first GUI at the display device while retaining the connection to the video conference via the portable device. The further display of the first GUI at the display device includes GUI information for rendering, at the display device, one or more conference participants, a whiteboard of the video conference, a presentation of the video conference, a text transcription of the audio of the video conference, or a chat room for the video conference.

(81) At **708**, the method **700** includes, based on the detected connection, displaying a second GUI that includes conference controls at the portable device. The client application displays the second GUI at the portable device while displaying the first GUI at the display device. In some implementations, the first GUI and the second GUI may form a switchable UI that is configured to switch between the first GUI and the second GUI based on the detected connection between the portable device and the display device or a detected disconnection of the display device from the portable device. In some implementations, second GUI may be automatically generated based on a current status of the video conference. For example, if someone is sharing their screen, the second GUI may be different than the second GUI of a one-on-one video conference.

(82) An aspect may include a method that includes displaying, at a portable device connected to a video conference, a first GUI that includes first video content of the video conference with a client application running at the portable device. The method may include detecting, during the video conference by the client application, a connection between the portable device and a display device. Based on the detected connection, the method may include causing, by the client application while retaining the connection to the video conference via the portable device, a further display of the first GUI at the display device instead of the portable device. Based on the detected connection, the method may include displaying, at the portable device while the first GUI remains displayed at the display device, a second GUI that includes second video content of the video conference.

(83) An aspect may include a device that includes a processor and a display. The processor may be configured to run a client application. The display may be configured to display a first GUI that includes first video content of a video conference within the client application. The processor may be configured to detect, during the video conference, a connection between the device and a display device. The processor may be configured to, based on the detected connection, cause a further display of the first GUI at the display device instead of the device. The processor may be configured to, based on the detected connection, retain the connection to the video conference. The processor may be configured to, based on the detected connection, display a second GUI that includes second video content of the video conference on the display while the first GUI remains displayed at the display device.

(84) An aspect may include a non-transitory computer-readable medium comprising instructions stored in a memory, that when executed by a processor, cause the processor to perform operations. The operations may include displaying, at a portable device connected to a video conference, a first GUI that includes first video content of the video conference within a client application running at the portable device. The operations may include detecting, during the video conference by the client application, a connection between the portable device and a display device. The operations may include, based on the detected connection, causing, by the client application while retaining the connection to the video conference via the portable device, a further display of the first GUI at the display device instead of the portable device. The operations may include, based on the detected connection, displaying, at the portable device while the first GUI remains displayed at the display device, a second GUI that includes second video content of the video conference.

(85) In one or more aspects, the second GUI may include controls for the video conference. In one or more aspects, the display device may be an external display that is not connected to the video conference. In one or more aspects, the further display of the first GUI at the display device may include graphical user interface information for rendering conference participants on the display device. In one or more aspects, the further display of the first GUI at the display device may include graphical user interface information for rendering a whiteboard of the video conference on the display device. In one or more aspects, the layouts of the first GUI and the second GUI may each be configurable by a user of the portable device. In one or more aspects, the client application may be used to configure a switchable interface configured to switch between the first GUI and the second GUI on the device based on the detected connection. In one or more aspects, the first GUI may be displayed on the display based on a determination that the device and the display device are disconnected. In one or more aspects, the display of the second GUI on the display may be terminated based on a determination that the device and the display device are disconnected. In one or more aspects, the first GUI may be displayed on the display. In one or more aspects, the further display of the first GUI may include GUI information for rendering a presentation of the video conference. In one or more aspects, the further display of the first GUI may include GUI information for rendering a text transcription of the audio of the video conference. In one or more aspects, it may be determined that the portable device and the display device are disconnected. In one or more aspects, the display of the second GUI at the portable device may be terminated. In one or more aspects, the first GUI may be displayed at the portable device. In one or more aspects, audio for the video conference may be obtained via a microphone of the portable device when displaying the first GUI at the display device and the second GUI is displayed at the portable device. In one or more aspects, video for the video conference may be obtained via a camera of the portable device when the first GUI is displayed at the display device and the second GUI is displayed at the portable device. In one or more aspects, the further display of the first GUI at the display device may include GUI information for rendering a chat room of the video conference on the display device. In one or more aspects, the further display of the first GUI at the display device may include GUI information based on a user preference. In one or more aspects, the further display of the first GUI at the display device may include GUI information based on a user preference that is learned using an ML model.

(86) The implementations of this disclosure can be described in terms of functional block components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes, routines, or other programming elements.

(87) Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc. Likewise, the terms “system” or “tool” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a combination of software and hardware. In certain contexts, such systems or mechanisms may be

understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.

(88) Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be a device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with a processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.

(89) Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media, and can include volatile memory or non-volatile memory that can change over time. The quality of memory or media being non-transitory refers to such memory or media storing data for some period of time or otherwise based on device power or a device power cycle. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.

(90) While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. A method comprising: displaying, at a portable device connected to a video conference, a first graphical user interface (GUI) that includes first video content of the video conference within a client application running at the portable device; detecting, during the video conference by the client application, a connection between the portable device and a display device; based on the detected connection: causing, by the client application while retaining the connection to the video conference via the portable device, a further display of the first GUI at the display device instead of the portable device; and displaying, at the portable device while the first GUI remains displayed at the display device, a second GUI that includes second video content of the video conference, wherein the second GUI is configured using a machine learning (ML) model that is trained using a data set of previous instances of display preferences.
2. The method of claim 1, wherein the second GUI includes controls for the video conference.
3. The method of claim 1, wherein the display device is an external display that is not connected to the video conference.
4. The method of claim 1, wherein the further display of the first GUI at the display device includes graphical user interface information for rendering conference participants on the display device.
5. The method of claim 1, wherein the further display of the first GUI at the display device includes graphical user interface information for rendering a whiteboard of the video conference on the display device.
6. The method of claim 1, wherein layouts of the first GUI and the second GUI are each configurable by a user of the portable device.
7. A device comprising: a processor configured to run a client application; and a display configured to display a first graphical user interface (GUI) that includes first video content of a video conference within the client application; the processor further configured to: detect, during the video conference, a connection between the device and a display device; and based on the detected connection: cause a further display of the first GUI at the display device instead of the device;

retain the connection to the video conference; and display a second GUI that includes second video content of the video conference on the display while the first GUI remains displayed at the display device, wherein the second GUI is configured using a machine learning (ML) model that is trained using a data set of previous instances of display preferences.

8. The device of claim 7, wherein the connection between the device and the display device is a wireless connection.

9. The device of claim 7, wherein the processor is further configured to: configure, using the client application, a switchable interface configured to switch between the first GUI and the second GUI on the device based on the detected connection.

10. The device of claim 7, wherein the processor is further configured to: display the first GUI on the display based on a determination that the device and the display device are disconnected.

11. The device of claim 7, wherein the processor is further configured to: terminate the display of the second GUI on the display based on a determination that the device and the display device are disconnected; and display the first GUI on the display.

12. The device of claim 7, wherein the further display of the first GUI includes GUI information for rendering a presentation of the video conference.

13. The device of claim 7, wherein the further display of the first GUI includes GUI information for rendering a text transcription of audio of the video conference.

14. A non-transitory computer-readable medium comprising instructions stored in a memory, that when executed by a processor, cause the processor to perform operations comprising: displaying, at a portable device connected to a video conference, a first graphical user interface (GUI) that includes first video content of the video conference within a client application running at the portable device; detecting, during the video conference by the client application, a connection between the portable device and a display device; based on the detected connection: causing, by the client application while retaining the connection to the video conference via the portable device, a further display of the first GUI at the display device instead of the portable device; and displaying, at the portable device while the first GUI remains displayed at the display device, a second GUI that includes second video content of the video conference, wherein the second GUI is configured using a machine learning (ML) model that is trained using a data set of previous instances of display preferences.

15. The non-transitory computer-readable medium of claim 14, the operations further comprising: determining that the portable device and the display device are disconnected; terminating the display of the second GUI at the portable device; and displaying the first GUI at the portable device.

16. The non-transitory computer-readable medium of claim 14, the operations further comprising: obtaining audio for the video conference via a microphone of the portable device when displaying the first GUI at the display device and displaying the second GUI at the portable device.

17. The non-transitory computer-readable medium of claim 14, the operations further comprising: obtaining video for the video conference via a camera of the portable device when displaying the first GUI at the display device and displaying the second GUI at the portable device.

18. The non-transitory computer-readable medium of claim 14, wherein the further display of the first GUI at the display device includes GUI information for rendering a chat room of the video conference on the display device.

19. The non-transitory computer-readable medium of claim 14, wherein the further display of the first GUI at the display device includes GUI information based on a user preference.

20. The non-transitory computer-readable medium of claim 14, wherein the further display of the first GUI at the display device includes GUI information based on a user preference that is learned using the ML model.
